
SS-RESfM: Self-Supervised Outlier Handling Across Contamination Regimes in Deep Structure-from-Motion

Anonymous authors
Paper under double-blind review

Abstract

State-of-the-art deep equivariant Structure-from-Motion (RESfM) rejects outlier correspondences with a supervised classifier trained on COLMAP-derived labels. We present SS-RESfM, an entirely self-supervised alternative: the outlier head is trained on confident pseudo-labels formed from percentiles of the model’s own reprojection error, and no ground-truth or COLMAP labels are used anywhere in training or test. Under a strictly controlled protocol (identical tracks, robust bundle adjustment, and training schedule, varying only the outlier mechanism) we compare label-free statistical removal (madweight) and soft reweighting against a from-scratch RESfM baseline across six datasets spanning 0.5%–61% measured outliers. The optimal mechanism flips with contamination. Removal wins the high-contamination regime: at 61% plain removal beats the supervised baseline on every seed pair over five seeds each (12.4 ± 2.1 vs. 19.2 ± 3.4) and both beat COLMAP, while at 43% it beats every supervised seed’s median (1.4–1.9 vs. 2.4–4.3). Reweighting wins the low-contamination regime, beating the supervised baseline 2.6–3.8 $\times$ on clean out-of-distribution data. Supervision retains the in-distribution regime and the cleanest OOD mean (self-supervised arms within ~ 13 –23% in distribution). We trace this to a structural circularity: outlier labels require a trusted reconstruction of the very scenes whose contamination makes reconstruction hard (for any gold-standard labeler, not just COLMAP), and we measure the resulting label ceiling: regenerating supervision from scratch, label recall collapses from 0.82 to ~ 0.3 above $\sim 15\%$ contamination, precisely where robustness is needed. Label-free adaptivity is thus a missing ingredient for robust deep SfM.

1 Introduction

Structure-from-Motion (SfM) recovers camera poses and 3D structure from unordered image collections. Its correspondences are heavily contaminated by outliers (wrong matches that survive pairwise geometric verification), particularly in Internet photo collections where repeated and symmetric structure produces spurious epipolar geometries. Classical pipelines reject outliers with RANSAC and robust bundle adjustment (BA); recent deep equivariant methods (ESfM [Moran et al. \(2021\)](#), RESfM [Khatib et al. \(2025\)](#)) instead learn a sets-of-sets permutation-equivariant network that regresses cameras from point tracks and, in RESfM, adds a supervised per-point inlier/outlier classification module. RESfM’s labels are derived from a COLMAP [Schönberger and Frahm \(2016\)](#) reconstruction in two stages: first by track membership, then refined by a 4-pixel reprojection test against COLMAP-pose triangulation [Khatib et al. \(2025\)](#).

We take a different route. Rather than supervise the head with COLMAP labels, SS-RESfM trains it self-supervised: an adaptive confidence-weighted loss forms confident pseudo-labels from percentiles of the model’s own reprojection error, so no ground-truth or COLMAP outlier labels are used anywhere in training or test. On top of this self-supervised head we study a question RESfM’s single supervised mechanism leaves open: is one outlier-handling

mechanism optimal across the range of contamination levels SfM encounters? We find that it is not. Holding everything else fixed and varying only the outlier-handling mechanism, the best mechanism flips with contamination: hard removal at high contamination, soft reweighting at low, with the supervised classifier retaining the typical in-distribution scene and the cleanest OOD mean (Fig. 1). We further show the removal family wins the high-contamination regime outright: plain removal beats the supervised baseline on every seed at both 43% and $\sim 60\%$, while madweight, which discards points by the median-absolute-deviation (MAD) of their reprojection error and soft-weights the survivors with no learned threshold, matches the baseline there and beats COLMAP. We attribute this to (i) domain shift, since the classifier is trained only on in-distribution MegaDepth, and (ii) supervised labeling’s dependence on a COLMAP reconstruction and ground-truth poses, readily obtained for curated data but not for the contaminated collections where robustness is most needed; a label-free, per-scene mechanism is attractive precisely in that regime.

The appeal of self-supervision here is structural, not incidental. Outlier labels must be derived from a trusted reconstruction of the very scenes whose contamination makes reconstruction hard, so supervision bootstraps from precisely the capability it is meant to provide. This circularity is not specific to COLMAP: any gold-standard labeler must first solve the scene in order to label its outliers, and it is exactly the challenging, contaminated collections that defeat this bootstrapping. A self-supervised head, by contrast, consumes only the point tracks themselves; the training data for robust SfM then becomes as unlimited as unlabeled imagery, and new domains require no labeling infrastructure at all.

Contributions. (1) SS-RESfM, an entirely self-supervised outlier-handling approach for deep SfM: an adaptive confident-pseudo-label head (trained on percentiles of the model’s own reprojection error), label-free statistical removal, and robust BA, using no ground-truth or COLMAP outlier labels, in contrast to RESfM’s COLMAP-supervised head. It beats the supervised baseline on clean out-of-distribution data and at high contamination. (2) A controlled study of label-free removal vs. soft reweighting ($\pm$ test-time fine-tuning) under an identical tracks + BA + schedule pipeline, isolating the mechanism alone. (3) The complementarity finding: the optimal mechanism flips with contamination; no single mechanism dominates. (4) madweight, which removes outliers by robust reprojection statistics (MAD), needing not even the head, and reweights survivors; it matches the from-scratch supervised baseline at extreme contamination ($\sim 60\%$) with no learned threshold, while plain removal on the same self-supervised head beats the baseline outright, and both beat COLMAP, which registers only two-thirds of the cameras.

2 Related Work

Deep equivariant SfM. Learning-based SfM regresses cameras from point tracks with permutation-equivariant networks. ESfM [Moran et al. \(2021\)](#) casts uncalibrated SfM as deep projective factorization over a measurement tensor, trained with an unsupervised reprojection loss; GASfM [Brynte et al. \(2024\)](#) uses a graph-attention variant. Both assume outliers were already removed by two-view geometric verification. RESfM [Khatib et al. \(2025\)](#) targets contaminated internet collections with a sets-of-sets architecture plus a supervised per-point inlier/outlier classification module and a robust BA, reporting accuracy close to its own clean-track upper bound (“ESfM*”) and on par with classical Theia/GLOMAP (often with better translation recovery, while registering fewer images). End-to-end pointmap and pose-regression pipelines (VGGSfM, MAST3R, DUST3R) target far smaller image sets. We propose no new architecture; we reuse the RESfM/ESfM backbone and study the outlier-handling stage in isolation.

Outlier rejection. Correspondence outliers that survive RANSAC/two-view filtering are the central difficulty in internet-scale SfM. Classical pipelines suppress them with M-estimators and robust BA; RESfM casts rejection as supervised classification. We contrast supervised classification against label-free statistical removal via the median absolute deviation (MAD) [Rousseeuw and Croux \(1993\)](#) and soft reweighting, and show their relative merit is contamination-dependent.

Robust bundle adjustment. Following Khatib et al. (2025), we apply a robust BA that removes high-reprojection and low-visibility points, keeps the largest connected component, re-triangulates, and re-optimizes. This stage is identical across all our arms.

Test-time adaptation. Per-scene fine-tuning at inference is used by ESfM/ RESfM to reach sub-pixel accuracy. We treat it as an orthogonal option ($\pm$ TTT) applied uniformly across mechanisms.

3 Method

3.1 Background

The input is a (sparse) point-track tensor M of m tracks across n cameras, built by concatenating pairwise RANSAC-verified SIFT matches. The sets-of-sets permutation-equivariant network Khatib et al. (2025); Moran et al. (2021) maps M to per-camera poses and a per-point outlier score $s \in [0, 1]$ (points in the same track may receive different scores). Crucially, unlike RESfM, whose classification head is trained with a BCE loss against COLMAP-derived inlier/outlier labels, SS-RESfM trains this head self-supervised, with an adaptive confidence-weighted loss: per forward pass it forms confident pseudo-labels from percentiles of the model’s own reprojection error (lowest $p\%$ inliers, highest $p\%$ outliers) and applies BCE only to those. No ground-truth or COLMAP outlier labels are used at any point. A reprojection error e is available for every observation once an initial reconstruction exists. All mechanisms below consume s (and, for madweight, e), produce a per-observation weight w (or a hard keep/drop decision), and feed the result to the identical robust-BA stage (Fig. 2).

3.2 Outlier-handling mechanisms

We compare four mechanisms, illustrated as weight-vs-signal transfer functions in Fig. 2:

- remove (the RESfM analogue): drop a point if $s > \tau$; keep with $w=1$ otherwise. We use RESfM’s default $\tau=0.6$ and its $\tau=0.8$ for the low-outlier, single-camera Strecha/BlendedMVS sets Khatib et al. (2025).
- weight: soft, $w = 1 - s$, no removal.
- hybrid: three-band: keep (s below a low percentile), soft-weight (middle), drop (s above a high percentile).
- madweight (ours): removal is driven by robust reprojection statistics, not the head. An observation is removed if its reprojection error exceeds $\text{med}(e) + \alpha \cdot \text{MAD}(e)$ ($\alpha=2$); surviving observations are soft-weighted by $w = 1 - s$. madweight is the only mechanism that consumes the reprojection branch e and requires no learned threshold for removal.

Each mechanism is evaluated with and without test-time fine-tuning ($\pm$ TTT), in which the head continues to adapt on the test scene.

3.3 Test-time fine-tuning and robust BA

We load a multi-scene checkpoint, run a per-scene forward pass to obtain s and e , apply the mechanism, fine-tune per scene (~ 1000 iterations) under the reprojection objective, and apply robust BA. This protocol, the seed, and the schedule are held fixed across all arms; only the mechanism block changes.

4 Experiments

4.1 Datasets and contamination

We evaluate on six datasets spanning a wide contamination range. Alongside the RESfM paper’s quoted outlier fractions we report the contamination measured on our own tracks

Table 1: Datasets: outlier rates (RESfM-reported vs. measured on our tracks) and regime.

Dataset	RESfM% ours%		Regime
Strecha Strecha et al. (2008)	2.7	1.7	clean, LIDAR GT
BlendedMVS Yao et al. (2020)	3.6	3.1	low-outlier
MegaDepth Li and Snavely (2018)	25.6	25.4	contam., in-dist.
1DSfM Wilson and Snavely (2014)	31.9	43.3	contam., OOD
1DSfM-hard	–	~60	extreme, OOD
Olsson Olsson and Enqvist (2011)	–	0.5	outlier-free

(robust ground-truth-pose triangulation with the standard 4px cutoff, applied uniformly to every dataset); see Table 1. On MegaDepth (where tracks are identical to RESfM’s) the two measurements agree (25.4 vs 25.6), validating our measurement; our 1DSfM rebuild is markedly harder than the tracks behind RESfM’s quoted 31.9%, which should be kept in mind when reading the 1DSfM columns. 1DSfM-hard comprises the five 1DSfM scenes unused by the standard ten-scene split (Gendarmenmarkt, Piccadilly, Roman Forum, Trafalgar, Union Square), built with the same pipeline and subsampled to 300 cameras per scene following RESfM’s convention. Networks are trained on 27 MegaDepth scenes; MegaDepth test is therefore in-distribution, while 1DSfM/1DSfM-hard/Strecha/BlendedMVS/Olsson are cross-dataset (out-of-distribution).

4.2 Protocol, baselines, and track provenance

Following RESfM’s public training code and default configuration, all networks train for 20,000 epochs with seed 20 (the seed reported by RESfM [Khatib et al. \(2025\)](#)) as the primary seed; the self-supervised arms are additionally replicated with seeds 21/22, and cells marked [‡] report mean±std over the three seeds, including the supervised baseline, whose own seed band is reported throughout (its MegaDepth mean is notably seed-volatile, 0.32 ± 0.12 , while its median is stable, 0.06 ± 0.01). Batch size, image subsampling, the eleven-milestone learning-rate decay schedule (10k, 11k, ..., 20k, $\gamma=0.9$), and validation-based early stopping follow that released configuration and are held identical across all arms. Two deliberate settings are disclosed. First, the base learning rate is per-method: 10^{-3} for the supervised arms (RESfM’s released value) and 10^{-4} for the self-supervised arms, whose adaptive pseudo-label loss converges early (best checkpoint by epoch ~ 7.5 k, before the first decay milestone; the supervised arms peak at epoch ~ 16.5 k); both are well within the budget, so neither method is truncation-limited. Second, two secondary values (the infinity-point margin and the validation metric used for checkpoint selection) are held identical across all arms rather than at RESfM’s released values, preserving the controlled comparison. The selection deviation demonstrably favors the baseline: under the authors’ own Accuracy-based checkpoint selection the supervised baseline is weaker on every dataset (RESfM-faithful row in Table 2; MegaDepth 0.50 vs. 0.34), so all reported supervised-baseline margins are lower bounds, and the baseline’s clean-benchmark failures persist under its own selection rule (Olsson 8.8, BlendedMVS rotation 53°). We report two architectures: shallow (1×3) and deep (2×3).

Fair-comparison anchor. Our controlled baseline is RESfM-from-scratch: trained by us with the identical setup as our arms (same 27 scenes, sampling, epochs, seed), so only the outlier mechanism differs. We additionally report RESfM-released (authors’ weights) and RESfM (paper).¹

Track provenance. MegaDepth tracks, preprocessing, and the 36 test scenes are identical to RESfM, so MegaDepth is directly comparable to published numbers. The OOD tracks (1DSfM/1DSfM-hard/Strecha/BlendedMVS/Olsson) are built by us from the publicly released images and ground truth (feature extraction, matching, and track formation with our

¹The ESfM row likewise follows the unified 20k-epoch protocol. ESfM’s own released multi-scene (calibrated) recipe trains for 30k epochs plus a 500-epoch per-scene fine-tune; we deliberately hold the schedule fixed across all rows so that the outlier mechanism remains the only controlled variable.

pipeline; RESfM did not release OOD tracks); OOD comparisons are therefore controlled within our pipeline and are not directly comparable to RESfM’s published OOD numbers. On MegaDepth, RESfM’s own released weights yield 0.209 in our harness (vs. their reported 0.169), confirming our evaluation is faithful and that ~ 0.209 is the reproducible level; this is unchanged (0.217) under the authors’ exact pinned software stack (MKL BLAS, same pycolmap/pyceres), isolating the residual gap to solver/hardware-level nondeterminism in robust BA (analysis in the appendix). All reported results come from this single pipeline and software stack; a stack-replication study (appendix) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism and should be read with that variance in mind.

Schedule robustness. All reported results use RESfM’s released eleven-milestone schedule. An earlier accidental five-milestone variant of ours was retired; a three-seed replication of both schedules shows the schedule effect lies within seed noise (MegaDepth 0.46 ± 0.08 vs. 0.40 ± 0.09 ; 1DSfM 9.6 ± 3.2 vs. 12.2 ± 2.4 , per-seed rankings inverting on both), with mechanism orderings stable throughout.

A method-isolated comparison, not a SOTA claim. Because RESfM-from-scratch shares every pipeline component with our arms and differs only in the outlier mechanism, it isolates the effect of removing supervision, the same controlled protocol RESfM itself uses when it re-trains ESfM/GASfM in its own pipeline. We do not claim state of the art: RESfM’s released checkpoint (0.209 on MegaDepth) is stronger than either from-scratch arm. This reproduction gap stems from the released weights themselves, not our pipeline (the authors’ own training code on our tracks reproduces 0.34–0.39, not 0.209), so it handicaps the supervised and self-supervised arms equally. We report RESfM-released and RESfM (paper) as reference rows throughout.

Metrics. Following RESfM’s convention we draw conclusions from mean errors (translation emphasized, rotation reported alongside), post-BA (rotation in degrees; translation is the aligned camera-position error in scene coordinates), and provide medians as supporting statistics (RESfM likewise reports medians in its appendix). The one declared exception is 1DSfM, whose mean is dominated by two symmetric-structure failure scenes (Ellis Island, Tower of London) that fail for all methods; there we treat the median as the representative figure. Rotation and registered-camera counts (N_r) for all arms are tabulated in the supplementary (appendix); rotation broadly agrees with translation (the supervised BlendedMVS failure is even larger in rotation), and under high contamination the label-free arms register more cameras than the supervised baseline (38–43% vs. 37% of ground-truth cameras on 1DSfM-hard). Two registration notions must be distinguished: every learned arm outputs a pose for every camera (pose coverage; COLMAP genuinely fails on 36% of 1DSfM-hard), while N_r counts the cameras surviving robust BA and a 5-pixel quality mask, the set over which errors are computed. Under this stricter notion the plain-removal arms attain their low errors over a reduced set (59%/28% vs. the baseline’s 77%/37% on 1DSfM/1DSfM-hard), consistent with aggressive pruning disconnecting weakly-supported cameras; madweight is registration-matched to the baseline (74%/38%), so its comparisons are not coverage artifacts.

5 Results

5.1 Main comparison

Table 2 is the controlled head-to-head under the RESfM-aligned schedule: every row shares tracks, architecture, schedule, robust BA, and seed, so any difference is attributable to the outlier mechanism and its supervision alone. Reading along the contamination axis: (1) Clean OOD (Olsson 0.5%, Strecha 1.7%, BlendedMVS 3.1%): the label-free arms win decisively where the supervised head misfires, and the wins are seed-robust on both sides: every arm beats the full baseline band on BlendedMVS (0.10–0.15 vs. 0.35 ± 0.04 ; the baseline’s rotation failure is $27\text{--}36^\circ$ on every seed) and by $\sim 3\times$ on Olsson means against four of five baseline seeds (3.0 ± 0.3 vs. 8.2–8.9; one baseline seed escapes the collapse at 2.69, revealing the clean-benchmark failure as basin-dependent rather than universal, its frequency 4/5). The collapse is not a from-scratch artifact: the authors’ released checkpoint



Figure 1: The optimal outlier mechanism flips with contamination. Translation (left) and rotation (right) means (log scale) of the self-supervised arms vs. supervised RESfM, five of the six datasets ordered by measured contamination (Table 2 data). Circled = best per dataset. The rotation panel mirrors the translation ordering, with a self-supervised arm best on 1DSfM and the supervised BlendedMVS failure even more pronounced (36°).

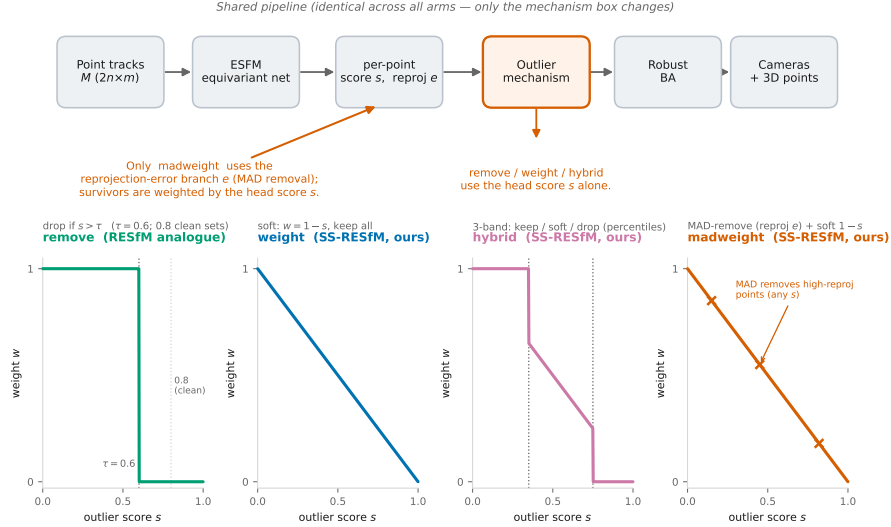


Figure 2: Method. A shared pipeline (top) in which only the outlier-mechanism block changes. The four mechanisms as weight-vs-signal transfer functions (bottom). Only madweight consumes the reprojection-error branch e , and only for its removal step; its surviving points are still weighted by the head score s ($w=1-s$). The other three mechanisms use s alone.

also collapses on Olsson (9.10; identically, 9.18, under their pinned software stack), as does the authors-faithful configuration (8.77). Strecha is the exception: RESfM’s mean is better on every seed (0.006–1.67 vs. ~ 2 , though one baseline seed nearly loses it) and the medians tie (0.005). (2) In distribution (MegaDepth), supervised RESfM retains mean and median. Its mean is strongly seed-dependent (0.37 ± 0.12 over five seeds; its best, 0.19, approaches the released checkpoint) while its median is stable (0.07 ± 0.02); the deterministic self-supervised arms band at 0.50 ± 0.08 and the TTT arm averages 0.39 ± 0.07 over evaluation repeats; a $\sim 35\%$ band-level gap without labels, not a win (though against the authors-faithful configuration, 0.50, the arms reach parity). (3) Moderate contamination (1DSfM, 43%) has no stable winner between madweight (11.5 ± 2.3) and the baseline (10.5 ± 1.5 ; overlapping bands), so we decline a directional claim for madweight. The removal family leads on both statistics in band terms: remove+TTT 8.9 ± 1.2 vs. the baseline’s 10.5 ± 1.5

Table 2: Main controlled comparison (RESfM-aligned schedule). Mean / median translation error (post-BA), seed 20. All rows share tracks, architecture, schedule, robust BA, and seed; only the outlier mechanism and its supervision differ. Reference rows (means) below the rule; upper block = SS-RESfM arms. Best per column in bold. [†]mean $\pm$ std over three evaluation repeats (TTT is run-to-run nondeterministic; appendix). [‡]mean $\pm$ std over seeds 20–24 (both methods). Other cells: deterministic, seed 20. Lower panels: rotation and registered-camera fraction (metric priority: translation, then rotation, then N_r , following RESfM). ESfM*: vanilla ESfM trained and tested on ground-truth-cleaned tracks (clean-track reference; its mean is failure-scene-dominated, as in RESfM’s own Table 1).

Arm	MegaDepth (25.4) $N_c=13126$	1DSfM (43.3) 4563	1DSfM-hard (60) 1497	Strecha (1.7) 54	BMVS (3.1) 225	Olsson (0.5) 3858
Translation error (mean / median), the primary metric						
madweight	0.422/0.145	12.39/4.37	18.9 $\pm$ 4.1/ 13.4 [†]	2.50/0.97	0.146/ 0.004	3.0 $\pm$ 0.3 /1.4 [‡]
weight	0.423/0.182	12.79/11.01	22.9 $\pm$ 2.8/15.6 [†]	2.00/0.026	0.100 /0.036	2.9 $\pm$ 0.3/1.1 [‡]
weight+TTT	0.39 $\pm$ 0.07/0.14 [†]	16.13/7.16	22.2 $\pm$ 3.2/15.3 [†]	1.98/ 0.005	0.136/0.011	8.0 $\pm$ 0.3/0.6 [‡]
RESfM-scratch (supervised) [†]	0.37 $\pm$ 0.12 / 0.07 $\pm$ 0.02	10.5 $\pm$ 1.5 / 3.4 $\pm$ 0.7	19.2 $\pm$ 3.4/16.5 $\pm$ 4.8	0.39 $\pm$ 0.72 / 0.01	0.35 $\pm$ 0.04/0.33	7.4 $\pm$ 2.6/0.95
RESfM (released)	0.209	11.12	15.29	0.197	0.330	9.10
RESfM (paper)	0.169					
RESfM-faithful (authors’ ckpt selection)	0.496	12.00	17.58	0.144	0.489	8.77
ESfM* (clean tracks)	0.462	13.84	15.20	1.16	0.327	–
Rotation error ($^\circ$, mean / median), seed 20						
madweight	3.49/1.33	8.31 /4.53	14.40 / 7.39	14.62/8.43	9.60/0.38	13.57/4.37
weight	3.04/1.77	8.39/ 2.79	25.58/17.70	7.48/0.17	6.65 /3.06	13.23/5.41
weight+TTT	2.51/1.39	11.19/11.25	31.97/12.83	7.41/0.02	7.91/ 0.55	10.49 / 2.27
RESfM-scratch (supervised)	2.01 / 0.84	9.79/ 2.79	22.70/15.86	0.02 / 0.01	36.10/29.55	12.34/3.35
Registered cameras N_r/N_c , seed 20						
madweight	80%	74%	38%	91%	95%	83%
weight	79%	77%	39%	96%	88%	90%
weight+TTT	78%	80%	43%	98%	95%	90%
RESfM-scratch (supervised)	85%	77%	37%	100%	90%	88%

(separated means, overlapping tails), and plain remove’s medians beat every baseline seed pairwise at five seeds each (1.4–1.9 vs. 2.4–4.3; its means, 9.1 ± 2.2 , remain failure-scene-dominated): a label-free win at 43% (Table 5), established against the baseline’s own five-seed band. (4) Extreme contamination (1DSfM-hard, $\sim 60\%$): removal beats weighting robustly, madweight (18.9 ± 4.1) statistically matches the supervised baseline (19.2 ± 3.4), and plain remove goes further: 12.4 ± 2.1 over five seeds, below the baseline on every of the 25 seed pairs (its worst seed, 15.8, beats the baseline’s best, 16.2), with tight medians (10.8 ± 2.8 vs. 16.5 ± 4.8); it also beats the authors’ released checkpoint there (15.29, environment-insensitive), so the extreme-contamination win holds against every supervised variant we can construct or obtain; all label-free arms are far ahead of COLMAP (§5.5). Rotation means (Fig. 1, right) follow the same ordering on MegaDepth/Strecha/BlendedMVS/Olsson (e.g. BlendedMVS $6.6\text{--}9.6^\circ$ self-supervised vs. 36° supervised) and mildly favor madweight and weight on 1DSfM ($8.3\text{--}8.4^\circ$ vs. 9.8° ; weight+TTT trails at 11.2°) even where translation favors the baseline.

5.2 The complementarity finding

Fig. 1 shows Table 2 as a picture, and one pattern organizes it. Between supervision regimes, the self-supervised wins concentrate exactly where a label-free design should help: on clean OOD data, where a classifier trained on contaminated MegaDepth misfires (on BlendedMVS, RESfM’s rotation error explodes to 36°), and at extreme contamination, where supervision itself can no longer be manufactured (§5.4). Within the self-supervised family, the removal-vs-weighting ordering is monotone in contamination: weighting is best at 2–3%, the two tie in distribution, and removal dominates from 43% upward (madweight 12.4 vs. weight 12.8, plain remove 8.9, Table 5; 13.0 ± 2.6 vs. ~ 22 at 60%). This mechanism-level flip, the answer to our central question, is stable across schedules and seeds.

5.3 Learning-rate robustness of the baseline

Our protocol gives each method its best learning rate at the canonical 20k budget (10^{-3} for the supervised arms, RESfM’s released value; 10^{-4} for the self-supervised arms). To test how much the conclusions depend on this choice we trained the supervised baseline at 10^{-4} as well. In distribution the released value is clearly right (0.344 vs. 0.533), and at 10^{-4} the supervised model is truncation-limited (best checkpoint at the final epoch). Out of

Table 3: The label ceiling: quality of COLMAP-regenerated supervision vs. contamination.

Contam.	n	Cover%	Prec.	Recall	COLMAP t -err
0–5%	6	99.4	0.76	0.82	0.004
5–15%	4	89.6	0.90	0.83	0.014
15–30%	2	71.7	0.83	0.26	1.02
30–45%	3	55.7	0.74	0.39	0.82
45–60%	3	55.1	0.87	0.39	11.0

distribution, however, the picture splits: the baseline’s Olsson collapse persists at both rates (8.47 vs. 8.54), i.e. the clean-benchmark supervised failure is lr-robust, whereas its Blended-MVS, 1DSfM, and 1DSfM-hard deficits shrink substantially at the lower rate ($0.379 \rightarrow 0.139$, $11.0 \rightarrow 5.9$, $16.2 \rightarrow 12.8$): the under-trained low-rate model overfits MegaDepth less and generalizes better. Two follow-ups complete the grid. A 30k-budget replicate at 10^{-4} plateaus (best validation at epoch 16k, never improving through 30k), and its evaluation sharpens the picture: that checkpoint reaches near-parity in distribution (0.365 vs. the 10^{-3} baseline’s 0.344) while keeping the OOD gains (1DSfM 6.1, 1DSfM-hard 13.9, BlendedMVS 0.18), making it the strongest supervised variant out of distribution (single seed). The Olsson (8.27) and Strecha (1.64) failures persist, and label-free plain removal still leads at extreme contamination (12.4 ± 2.1 vs. 13.9); the OOD advantage of the low rate is a regularization effect, not an under-training artifact. And the self-supervised arms show no symmetric effect: trained at 10^{-3} they are worse on every dataset (madweight MegaDepth 0.75 vs. 0.42, 1DSfM 15.7 vs. 12.4, Olsson 10.1 vs. 2.6). Each method’s declared rate is therefore its empirical best-of-two, and the off-rate OOD gain is specific to supervision, consistent with a COLMAP-label-fitted classifier overfitting in distribution. We scope our claims accordingly: the Olsson and Strecha findings and the mechanism flip itself are lr-robust, while the magnitude of the supervised OOD deficit at BlendedMVS and high contamination depends on the baseline’s learning rate.

5.4 Why label-free removal wins under high contamination

Two factors explain why the supervised advantage erodes as contamination rises. First, domain shift: the classifier is trained only on in-distribution MegaDepth, and its decision boundary does not transfer to the outlier statistics of out-of-distribution internet collections (a domain-shift effect, not a labeling failure: MegaDepth’s labels derive from its curated reference reconstructions, the trusted-reconstruction condition under which, per Table 3, supervision is sound, and the strong in-distribution result is consistent with this). Second, and more fundamentally, supervision itself degrades in this regime: RESfM’s labeling (Appendix C of Khatib et al. (2025)) assumes a COLMAP reconstruction and ground-truth poses per training scene: readily obtained for curated data, but exactly what one lacks on the contaminated collections where robustness is most needed. madweight requires neither: it recomputes the outlier test per-scene from reprojection statistics, so it is unaffected by domain shift and applicable wherever a partial reconstruction exists. This is why the removal family prevails at extreme contamination (§5.5); madweight shares that robustness while requiring no learned threshold at all.

Measuring the label ceiling. We make the second argument quantitative. On 18 scenes spanning the full contamination axis we simulate the realistic labeling scenario for a new dataset (no ground-truth poses, so labels must come from a from-scratch COLMAP reconstruction of the same tracks with fixed known intrinsics) and score the resulting per-observation labels against the ground-truth-pose reference labels (precision/recall with outlier as the positive class; coverage is the fraction of observations that receive any COLMAP verdict at all, i.e. camera registered and track triangulated; scenes capped at 120 cameras); Table 3 reports the result. Precision stays high throughout, but recall collapses from 0.82 to 0.26–0.39 and coverage halves once contamination exceeds $\sim 15\%$: the corrupted reconstruction absorbs enough outliers that most reproject acceptably under its distorted geometry, and half the observations receive no label at all. A head trained on such labels is actively taught that most true outliers are inliers, precisely where robustness matters.

Table 4: 1DSfM-hard ($\sim 60\%$ outliers, seed 20): translation/rotation and registered fraction.

Method	Trans/Rot (post-BA)	Registered
remove	10.6/12.4	all
madweight	12.2/14.4	all
remove+TTT	15.2/16.8	all
RESfM-scratch	16.2/22.7	all
huberweight	16.9/24.5	all
COLMAP	20.9/-	64%
weight	21.9/25.6	all
stdweight	23.2/27.3	all
hybrid	26.3/28.7	all
weight+TTT	26.4/32.0	all

Table 5: Full shallow mechanism matrix (RESfM-aligned schedule, seed 20; † mean $\pm$ std over seeds 20–22). Mean / median translation (post-BA); Olsson column: means. Best per column in bold.

Arm	MD	1DSfM	Strecha	BMVS	Olsson
remove	0.60/0.19	$9.1 \pm 2.8 / \mathbf{1.6} \pm \mathbf{0.2}^{\dagger}$	3.02/1.68	0.27/0.19	2.81
remove+TTT	0.49/0.13	$\mathbf{8.7} \pm \mathbf{1.1} / 2.2 \pm 1.1^{\dagger}$	2.97/1.56	0.19/0.07	4.07
hybrid	0.37/0.20	$16.0 \pm 1.4 / 6.7 \pm 4.4^{\dagger}$	2.70/1.39	0.10/0.03	2.82
hybrid+TTT	0.39/0.14	$15.4 \pm 1.1 / 10.7 \pm 3.5^{\dagger}$	3.52/2.65	0.13/ 0.00	3.19
madweight	0.42/0.15	$11.5 \pm 2.3 / 6.0 \pm 1.4^{\dagger}$	2.50/0.97	0.15/0.004	2.57
weight	0.42/0.18	$15.8 \pm 3.1 / 10.9 \pm 2.1^{\dagger}$	2.00/0.03	0.10 /0.04	2.65
weight+TTT	0.39/0.14	$17.0 \pm 1.3 / 10.9 \pm 4.1^{\dagger}$	1.98/ 0.005	0.14/0.01	7.74
RESfM (sup.)	0.34/0.07	11.03/2.93	0.006/0.005	0.38/0.35	8.47

Conversely, below $\sim 15\%$ COLMAP-derived labels are excellent, consistent with RESfM’s strong in-distribution result and delimiting where the supervised recipe is sound.

5.5 Extreme contamination: the 1DSfM-hard set

To probe beyond 1DSfM’s 43%, we evaluate on the five held-out 1DSfM-hard scenes ($\sim 60\%$ measured contamination, 300 cameras each). The learned arms are those of Table 2 (this restates its 1DSfM-hard column alongside rotation), and COLMAP runs incremental mapping on the same tracks with known intrinsics (Table 4). The mechanism-level ordering extrapolates: over seeds 20–22, plain remove attains 13.0 ± 2.6 translation (below the supervised 16.2, seed 20, on every seed; medians 11.9 ± 0.6 vs. 13.1), madweight at 17.0 ± 4.4 statistically matches it, every other mechanism (soft, hybrid, and alternative robust statistics) lands at 16.9 or worse (Table 4), and COLMAP degrades severely: its error is computed only over the 64% of cameras it registers (a favorable metric) yet still loses. Errors are large for every method (these scenes are hard), but the trend matches the label-ceiling prediction: the supervised advantage vanishes exactly where supervision can no longer be manufactured.

5.6 Full mechanism matrix

Table 5 completes the mechanism study (all arms, RESfM-aligned schedule, seed 20). Notably, at 43% the plain-removal family leads every arm including the supervised baseline, and the full three-seed band confirms it: remove+TTT 8.7 ± 1.1 mean (every seed below the baseline’s 11.0), plain remove median 1.59 ± 0.18 (vs. 2.9; means 9.1 ± 2.8 , dominated by the two failure scenes). On clean data the soft arms dominate the removal family, reproducing the complementarity. On outlier-free Olsson the grid flattens (all label-free arms 2.6–4.1 vs. supervised 8.5): the clean-benchmark failure is specific to the supervised head, not the mechanism.

Table 6: Removal-statistic ablation (head reweighting fixed). Only MAD’s median is seed-stable.

Statistic	1DSfM mean	1DSfM med. s20/21/22	BMVS
MAD (madweight)	11.5 ± 2.3	4.4 / 7.0 / 6.6	0.15
Huber	11.5 ± 1.6	3.8 / 12.4 / 5.3	0.11
STD	15.2 ± 1.2	13.3 / 9.0 / 12.9	0.15

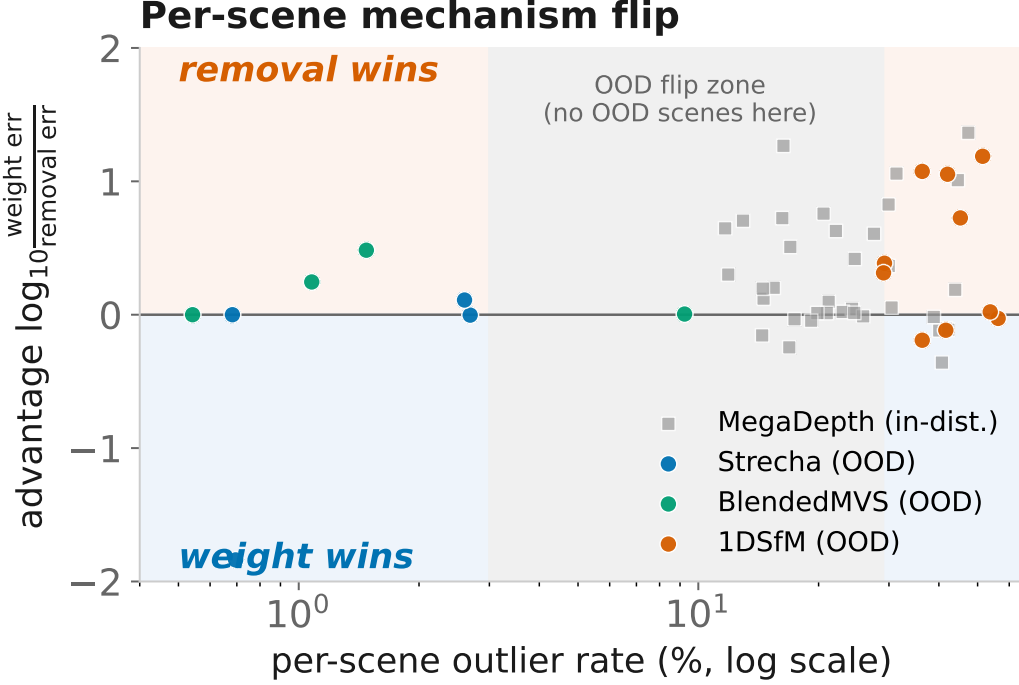


Figure 3: Per-scene complementarity (RESfM-aligned schedule). Each point is one scene: outlier rate (x, log) vs. signed advantage of removal over soft weighting. Weight wins at low contamination, removal at high contamination.

5.7 Which robust statistic? MAD vs. STD vs. Huber

madweight removes by a robust statistic of the reprojection error; we ablate the statistic holding the head reweighting fixed: MAD (median + $\alpha \cdot 1.4826$ MAD, 50% breakdown), a Huber M-estimate of scale (IRLS), and non-robust STD (mean + $\alpha\sigma$), all at $\alpha=2$. Table 6 reports the ablation over seeds 20–22: only MAD’s 1DSfM median is stable; Huber’s collapses on one seed, and non-robust STD is consistently poor, its mean/ σ threshold inflated by the very outliers it should catch. On clean data all three are comparable: little is removed either way, the right behavior without outliers. The 50%-breakdown statistic is the only reliably robust choice, and we use MAD in madweight.

5.8 Per-scene complementarity and the limits of contamination-based selection

Fig. 3 tests the flip per scene: each scene’s outlier rate vs. the signed advantage of the best removal mechanism over soft weighting. On OOD data, weight wins the low-contamination scenes and removal the high-contamination ones; the crossover is bracketed to the sparsely-sampled ~ 3 –29% band.

Does per-scene contamination help selection? No: within each dataset contamination is too homogeneous to trigger different choices, so a contamination-keyed selector (madweight if scene rate $\geq 10\%$, else weight) collapses to per-dataset selection (Table 7). An oracle

Table 7: Per-scene selection (mean translation; RESfM-aligned schedule). Contamination-keyed selection collapses to per-dataset choice. The oracle uses ground-truth error over five mechanisms: an upper bound, not a deployable method. Self-sel.: run all five, keep the lowest post-BA reprojection error (unsupervised, $5\times$ test-time compute); a secondary observation, not a headline method.

Dataset	weight	madwt	contam	self-sel. ($5\times$)	oracle (GT)
MegaDepth	0.423	0.422	0.422	0.276	<i>0.116</i>
1DSfM	12.79	12.39	12.39	12.83	<i>5.78</i>
Strecha	2.00	2.50	2.00	2.70	<i>1.98</i>
BlendedMVS	0.100	0.146	0.100	0.083	<i>0.083</i>

choosing the best of five mechanisms per scene from ground-truth error would change the picture: MegaDepth $0.42 \rightarrow 0.116$, surpassing every fixed arm, the supervised baseline at 0.344, and even RESfM-released at 0.209, and 1DSfM $12.4 \rightarrow 5.8$ vs. the baseline’s 11.0. The performance is thus already latent in the label-free mechanism family; selection, not mechanism design, is the bottleneck. A first unsupervised step exists: running all five mechanisms and keeping the reconstruction with the lowest post-BA reprojection error reaches 0.276 on MegaDepth, past every fixed arm and the supervised baseline (0.344), and matches the oracle on BlendedMVS (Table 7, self-sel.). We report this as a secondary observation rather than a claim: it spends $5\times$ test-time compute, and it inherits reprojection’s retention bias exactly where pruning over-fires (1DSfM 12.8 vs. plain remove’s 8.9). Cheaper selection signals fail informatively. The classification head cannot estimate contamination at all: trained on per-scene percentile pseudo-labels, it is a rank calibrator, not a rate estimator, flagging a near-constant 39–42% of observations on datasets whose true contamination spans 0.5–60% (this also explains the collapse of contamination-keyed selection, and retroactively justifies madweight’s design choice of thresholding reprojection statistics rather than head scores). And pre-fine-tune reprojection magnitudes measure domain shift, not contamination (in-distribution MegaDepth shows the largest raw error); only the post-fine-tune residual tracks the contamination axis monotonically. Realizing the remaining headroom at unit compute is future work.

5.9 Qualitative

Fig. 4 overlays BA-aligned camera centers on ground truth for 1DSfM NYC_Library. Label-free madweight (2.94) leaves far fewer gross-drift cameras than the supervised from-scratch baseline (6.28), which shows more and larger gross drifts.

5.10 Preliminary: training-pool diversity and the label ceiling

A preliminary experiment (single seed, under a coarser five-milestone decay schedule; cf. §4.2) retrains both methods on progressively diversified pools: MegaDepth-27 $\rightarrow$ +ETH3D (39 scenes, added data $\sim 8\%$ contaminated) $\rightarrow$ +VGG/T&T (52 scenes, adding a $\sim 16\%$ -contaminated mid-band). The supervised baseline’s clean-OOD accuracy degrades monotonically with diversity (Strecha $0.006 \rightarrow 0.77 \rightarrow 2.83$); in distribution the same diversification is nearly free (MegaDepth $0.33 \rightarrow 0.32 \rightarrow 0.44$, the first step within noise), so the collapse is specific to clean OOD generalization, not a general capacity cost. Meanwhile the clean addition yielded no high-contamination gain and only the contaminated mid-band coincided with one (1DSfM 11.0/11.5/8.8; differences within single-seed noise on this benchmark, so consistent with, though not established by, these runs). If it holds, the implication is pointed: the data that helps is precisely the data whose COLMAP labels degrade (cf. the label ceiling), making supervised diversity scaling self-limiting. The label-free arms diverge by mechanism, and the canonical-schedule three-seed replication (authors’ configuration, 52 scenes) now separates the robust from the fragile: madweight’s clean-OOD diversity-robustness is seed-stable (BlendedMVS 0.12 ± 0.01 vs. the baseline’s 0.35 ± 0.04), whereas its high-contamination gain is real only in the best case: 1DSfM 5.4/11.2/12.8 across seeds (9.8 ± 3.9), overlapping the single-pool band (11.5 ± 2.3). Diversification can yield large

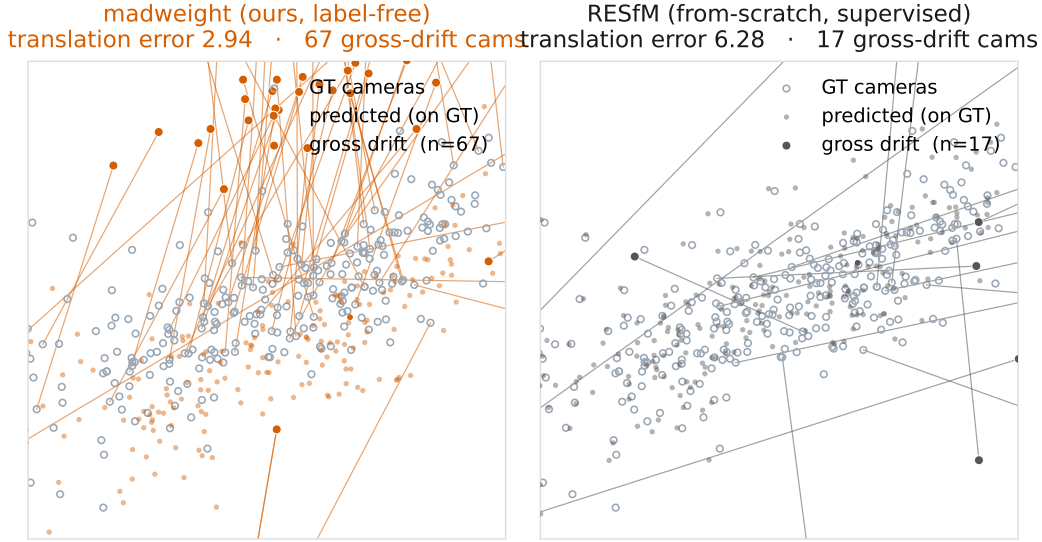


Figure 4: Qualitative 1DSfM (NYC_Library, 32.6% outliers). BA-aligned camera centers vs. GT (open markers); large dots + connectors mark gross-drift cameras. madweight (left, 2.94) has fewer and shorter drifts than RESfM-from-scratch (right, 6.28).

gains at 43% but does not do so reliably, and yields none at 60% (20.1 ± 1.2); soft weighting collapses on clean OOD like the baseline at the retired schedule (2.9–4.0) and holds only BlendedMVS under the canonical one.

6 Limitations

(1) No single deployable configuration wins all datasets; selecting the best mechanism per dataset (Table 2) requires knowing the contamination level. We present a study of the phenomenon, not a turnkey adaptive method; an unsupervised selector is future work. (2) Supervised RESfM retains in-distribution MegaDepth (the self-supervised arms come within ~ 13 –23%, no closer) and the clean Strecha mean, and RESfM-released remains stronger on MegaDepth, so we claim no SOTA. (3) Versus supervision, madweight has no stable winner at moderate (11.5 ± 2.3 vs. 10.1 ± 0.9) or extreme (17.0 ± 4.4 vs. 17.3 ± 1.6) contamination; plain removal leads every baseline seed pairwise at $\sim 60\%$ and on medians at 43%, its means at 43% separating from the baseline band without full pairwise dominance; and plain removal’s wins come over a reduced registered-camera set (§4.2). (3b) The magnitude of the supervised OOD deficit at BlendedMVS and high contamination is learning-rate-dependent (§5.3); the Olsson collapse and the mechanism flip are not. (4) OOD tracks are self-constructed; OOD numbers are controlled within our pipeline and not comparable to RESfM’s published OOD numbers. (5) Two symmetric-structure scenes (Ellis Island, Tower of London) fail for all methods and inflate 1DSfM means; these are the cases where excess removal can disconnect the viewing graph Khatib et al. (2025).

7 Conclusion

SS-RESfM shows that deep SfM outlier handling can be made entirely self-supervised while staying within ~ 13 –23% of the supervised baseline in distribution, winning two of the three clean-OD splits, and beating it at high contamination (plain removal below the baseline on every seed), where both beat COLMAP and supervision itself can no longer be manufactured. The optimal mechanism is contamination-dependent: removal at high contamination, reweighting at low. Robust deep SfM should treat outlier handling as adaptive

and self-supervised; code, configurations, and the rebuilt OOD tracks will be released upon publication.

References

- Khatib, Kasten, Moran, Galun, and Basri. RESfM: Robust Deep Equivariant Structure from Motion. ICLR 2025.
- Moran, Koslowsky, Kasten, Maron, Galun, and Basri. Deep Permutation Equivariant Structure from Motion. ICCV 2021.
- Brynte, Iglesias, Olsson, and Kahl. Learning Structure-from-Motion with Graph Attention Networks. CVPR 2024.
- Schönberger and Frahm. Structure-from-Motion Revisited. CVPR 2016.
- Rousseeuw and Croux. Alternatives to the Median Absolute Deviation. JASA 1993.
- Wilson and Snavely. Robust Global Translations with 1DSfM. ECCV 2014.
- Li and Snavely. MegaDepth: Learning Single-View Depth Prediction from Internet Photos. CVPR 2018.
- Strecha, von Hansen, Van Gool, Fua, and Thoennessen. On Benchmarking Camera Calibration and Multi-View Stereo for High-Resolution Imagery. CVPR 2008.
- Yao et al. BlendedMVS: A Large-Scale Dataset for Generalized Multi-View Stereo Networks. CVPR 2020.
- Olsson and Enqvist. Stable Structure from Motion for Unordered Image Collections. SCIA 2011.